

Paper II - Urology (MCh Urology, Summer 2015)

Contents

1. II-1. Medical management of BPH and role of surgical modalities (LAQ)
2. II-2. Metabolic evaluation in urinary calculus disease (LAQ)
3. II-3a. Management of priapism (SAQ)
4. II-3b. Renal tubular acidosis (SAQ)
5. II-3c. Millin's prostatectomy (SAQ)
6. II-3d. Energy sources in laparoscopic urology (SAQ)
7. II-3e. Role of vascular grafts in urology (SAQ)
8. II-3f. Transrectal ultrasonography (SAQ)

II-1. Describe medical management of BPH and role of various surgical modalities in its management.

Type: LAQ (25 marks)

Definition

- **BPH (benign prostatic hyperplasia)** = histological; **BPO (benign prostatic obstruction)** → **LUTS (lower urinary tract symptoms)**: storage, voiding, post-micturition.
- Drug targets: static (bulk, DHT (dihydrotestosterone)) → 5-ARIs (5 α -reductase inhibitors); dynamic (α 1A tone) → α -blockers.

Evaluation

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35. FVC (frequency-volume chart), 3 days.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)** (if life expectancy >10 y); optional Qmax (maximum flow rate; <10 mL/s suggests obstruction), PVR (post-void residual).
- **Before surgery** (AUA (American Urological Association)): prostate size/shape; pressure-flow study if diagnosis uncertain.

Medical management

1. Watchful waiting – mild/non-bothersome: fluid timing, ↓ caffeine/alcohol.

2. α 1-blockers – first line, bothersome LUTS; act within days.

Drug	Dose	Note
Tamsulosin	0.4 mg OD (once daily)	α 1A/ α 1D selective
Alfuzosin	10 mg OD	Least ejaculatory dysfunction
Silodosin	8 mg OD	Highest anejaculation
Doxazosin/terazosin	Titrate	Non-selective; orthostatic hypotension

- IPSS ↓ ~30–40%; **no ↓ volume or AUR (acute urinary retention)/surgery risk**.
- Side effects: postural hypotension, dizziness, ejaculatory dysfunction.
- **IFIS (intraoperative floppy iris syndrome)** – strongest with tamsulosin; **defer starting α -blocker if cataract surgery planned**.

3. 5-ARIs – finasteride 5 mg (type 2); dutasteride 0.5 mg (types 1 + 2).

- For **enlarged prostate** (AUA: PSA >1.5 ng/mL or >30 mL; EAU (European Association of Urology): >40 mL).
- Volume ↓ ~**20-25%** over 6-12 months; **PSA halved** (×2 after 6 months); AUR/surgery ↓ ~50%.
- Side effects: ↓ libido, ED (erectile dysfunction), gynaecomastia; ↑ high-grade cancer detection → FDA (Food and Drug Administration) warning (2011).

4. Combination

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: doxazosin + finasteride ↓ progression **66%** vs placebo (doxazosin 39%, finasteride 34%).
- **CombAT (Combination of Avodart and Tamsulosin; 4-yr)**: dutasteride + tamsulosin, volume ≥30 mL; AUR/surgery ↓ ~**66% vs tamsulosin**.
- Consider stopping α-blocker after 6-12 months.

5. Storage – antimuscarinics (caution **PVR >250-300 mL**, cognitive effects) or β3-agonists (mirabegron 25-50 mg).

6. Tadalafil 5 mg OD – LUTS ± ED; no nitrates.

7. Nocturnal polyuria – low-dose desmopressin; **serum sodium** at baseline, first week, periodically.

8. Phytotherapy – AUA: not recommended.

LUTS → IPSS, DRE, urinalysis, PSA, FVC
Mild → watchful waiting
Voiding → α-blocker (± 5-ARI if >30-40 mL)
Storage persists → antimuscarinic/β3 (check PVR)
Nocturnal polyuria → desmopressin
Failure/complications → surgery (size-based)

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection), refractory haematuria, bladder stones, renal insufficiency from BPO; LUTS failing medical therapy.

Modality	Size (AUA)	Pros	Cons
TURP (transurethral resection of prostate)	30-80 mL	Reference standard	Bleeding, retrograde ejaculation ~65-75%, TUR syndrome, stricture
TUIP (transurethral incision of prostate)	≤30 mL	Less retrograde ejaculation	Higher retreatment
PVP (photoselective vaporization)	30-80 mL	Anticoagulated patients	No histology
HoLEP (holmium laser enucleation)	Size-independent	Lowest retreatment, least bleeding	Learning curve
Simple prostatectomy	>80-100 mL	Complete removal	Morbidity, transfusion
Aquablation	30-80; consider 80-150 mL	Ejaculation preserved	Bleeding
Rezūm (water vapour)	30-80 mL	Day care, treats median lobe	Delayed relief
PUL (prostatic urethral lift)	30-80 mL, no median lobe	Sexual function preserved	Higher retreatment
PAE (prostatic artery embolisation)	Larger glands	No general anaesthesia	Less effective than TURP

TUR syndrome (monopolar, 1.5% glycine): dilutional hyponatraemia, fluid overload, glycine toxicity (confusion, bradycardia, visual disturbance); risk ↑ resection >60–90 min. Treat: stop surgery, diuretics, 3% saline if severe. **Bipolar TURP (saline)** virtually eliminates it.

Landmark trials

- **GOLIATH**: PVP **non-inferior** to TURP.
- **WATER (Waterjet Ablation Therapy for Endoscopic Resection of prostate tissue)**: Aquablation non-inferior to TURP, less anejaculation; **WATER II**: 80–150 mL.

Complications

- Bleeding, TUR syndrome, sepsis; retrograde ejaculation, transient incontinence, stricture, bladder neck contracture.

Follow-up

- IPSS at 4–12 weeks (α -blocker), 3–6 months (5-ARI), then yearly.

Recent advances

- **Optilume BPH** (paclitaxel-coated balloon) FDA-approved 2023.
- Office MISTs (minimally invasive surgical therapies) preserve ejaculation.

Diagram to draw: Size-based algorithm (≤ 30 TUIP $\rightarrow$ 30–80 TURP/PVP $\rightarrow$ >80 enucleation/simple prostatectomy).

II-2. Discuss metabolic evaluation in urinary calculus disease.

Type: LAQ (25 marks)

Introduction

- Recurrence ~50% at 5–10 years without prevention → find correctable risk factors.
- **Supersaturation** → nucleation → growth → aggregation. Promoters: Ca, oxalate, uric acid, low volume. Inhibitors: citrate, magnesium, pyrophosphate, Tamm-Horsfall protein.
- **RSS (relative supersaturation)** from 24-h urine guides therapy (target <1).

Who needs it

- **Basic - every stone former; comprehensive (24-h urine)** - high-risk, recurrent, interested first-time formers.
- **High-risk (EAU (European Association of Urology))**: early onset/children, family history, recurrent, solitary kidney, bilateral/large burden; brushite, uric acid, infection, cystine stones; **primary hyperparathyroidism**, GI (gastrointestinal) disease/bariatric surgery; cystinuria, primary hyperoxaluria, **RTA (renal tubular acidosis) type 1**; drugs (topiramate, acetazolamide, indinavir); medullary sponge kidney, calyceal diverticulum, horseshoe kidney.

Basic evaluation (all)

- **History**: age at onset, episodes, family history, diet (fluid, salt, animal protein, oxalate), drugs, bowel disease, gout.
- **Imaging**: NCCT (non-contrast computed tomography) - uric acid <500 HU (Hounsfield units).
- **Serum**: creatinine, **calcium, uric acid, bicarbonate** (low → RTA), phosphate; **PTH (parathyroid hormone)** if Ca high-normal/raised.
- **Urine: pH** (>6.5–7 → RTA/infection; <5.5 → uric acid), crystals (hexagonal = cystine; coffin-lid = struvite), culture (urease organisms).
- **Stone analysis** - first episode and recurrence on therapy: **infrared spectroscopy or X-ray diffraction**.
- **Cystine screen**: cyanide-nitroprusside test → confirm with 24-h cystine.

Comprehensive - 24-h urine

- **Two collections** (AUA (American Urological Association) one or two; EAU two), usual diet, ≥2–4 weeks after stone event, stone- and infection-free.
- Adequacy: **urinary creatinine** (men ~20–25, women ~15–20 mg/kg/d).

Parameter	Desirable	Abnormality
Volume	>2.5 L	<2 L - commonest
Calcium	<250 mg (F), <300 mg (M), or <4 mg/kg	Hypercalciuria
Oxalate	<40–45 mg	Hyperoxaluria
Citrate	>320 mg	Hypocitraturia
Uric acid	<750 mg (F), <800 mg (M)	Hyperuricosuria
Sodium	<100 mmol/d	↑ Ca excretion
pH	5.8–6.2	<5.5 uric acid; >6.5–7 CaP (calcium phosphate)/RTA
Magnesium	>70 mg	Hypomagnesiuria (GI)

Parameter	Desirable	Abnormality
Cystine	Very low	Cystinuria (> 400 mg/d)

Diagnostic categories

1. **Hypercalciuria** – absorptive, renal leak (↑ PTH, normal serum Ca), resorptive (primary hyperparathyroidism – ↑ Ca, ↑ PTH); thiazide works for all idiopathic.
2. **Hyperoxaluria** – dietary (45–60 mg/d); **enteric** (fat malabsorption: fatty acids bind Ca → free oxalate absorbed in colon; + hypocitraturia, low volume); **primary** (PH (primary hyperoxaluria); often >75–100 mg/d; genetic testing).
3. **Hypocitraturia** – idiopathic, **dRTA (distal RTA)**, chronic diarrhoea, hypokalaemia.
4. **Hyperuricosuria** – promotes **CaOx (calcium oxalate)** stones. **Uric acid stones: urine pH <5.5** (metabolic syndrome, diabetes).
5. **dRTA** – urine pH >5.5–6, hypocitraturia, CaP stones; incomplete form → **ammonium chloride test** (0.1 g/kg; pH fails to fall <**5.5**).
6. **Cystinuria** – autosomal recessive; COLA (cystine, ornithine, lysine, arginine); aim <**250 mg/L**.
7. **Infection stones** – urease organisms, pH >7, struvite.

Stone → basic evaluation

Low risk → general advice (output >2.5 L, ↓ Na/protein, normal Ca)

High risk/recurrent → two 24-h urines → specific abnormality

→ targeted therapy → repeat 24-h urine ~6 months → yearly

Treatment by abnormality

Abnormality	Diet	Drug
Hypercalciuria	Na <2.3 g/d, normal Ca, ↓ animal protein	Thiazide (HCTZ (hydrochlorothiazide) 25–50 mg, chlorthalidone 25 mg, indapamide 2.5 mg) + K citrate
Primary hyperparathyroidism	–	Parathyroidectomy
Hypocitraturia, dRTA	↑ fruit/vegetables	K citrate 30–60 mEq/d
Enteric hyperoxaluria	Low fat/oxalate, Ca with meals	Ca citrate, cholestyramine
PH1	High fluid	Pyridoxine, lumasiran ; liver ± kidney transplant
Hyperuricosuria	↓ purine	Allopurinol 100–300 mg
Uric acid stones	–	K citrate to pH 6.0–6.5
Cystinuria	Fluid >3–4 L	Alkalinise pH 7.0–7.5 ; tiopronin
Infection	–	Complete clearance, antibiotics

Special groups

- **Children: all need full evaluation**; spot urine ratios (to creatinine) if 24-h impractical; screen cystinuria, PH, RTA.
- **Pregnancy**: defer 24-h urine until **after delivery/lactation** (physiological hypercalciuria); avoid thiazides, allopurinol, tiopronin.

Follow-up

- **Repeat 24-h urine within 6 months**, then annually; drug toxicity checks; periodic imaging.

Recent advances

- **RNAi (ribonucleic acid interference) for PH1** (lumasiran, nedosiran).
- **NOSTONE (2023)**: hydrochlorothiazide did not ↓ recurrence vs placebo.

Diagram to draw: Supersaturation zones + evaluation algorithm.

II-3a. Management of priapism

Type: SAQ (10 marks)

Definition

- Full or partial erection **>4 hours** beyond or unrelated to sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2021).

Classification

	Ischaemic (low-flow)	Non-ischaemic (high-flow)	Stuttering
Mechanism	Veno-occlusive, compartment syndrome	Cavernous artery-sinusoidal fistula	Recurrent ischaemic
Cause	ICI (intracavernosal injection), SCD (sickle cell disease), antipsychotics	Perineal trauma	SCD
Penis	Rigid, painful	Partially rigid, painless	Painful
Emergency	Yes	No	Prevent progression

Diagnosis

- History (duration, drugs, trauma, SCD); glans soft in ischaemic.
- Corporal blood gas** - ischaemic: **pO₂ <30, pCO₂ >60 mmHg, pH <7.25** (normal arterial: >90, <40, 7.40).
- Colour Doppler**: absent flow (ischaemic); fistula (non-ischaemic).
- Blood count, haemoglobin electrophoresis, toxicology.

Ischaemic priapism

Dorsal penile block → aspiration (16–18G) ± saline irrigation
→ Phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to ~1 h (monitor blood pressure/pulse)
→ Failure → distal shunt → proximal shunt
→ >36 h / failed shunts → immediate penile prosthesis

- Distal shunts**: Winter (Tru-cut needle), Ebbehøj, T-shunt, Al-Ghorab (open); corporal tunnelling.
- Oral therapy alone not recommended.
- Proximal**: Quackels (cavernoso-spongiosal), Grayhack (cavernoso-saphenous).
- Ischaemia >24–48 h → fibrosis, ED (erectile dysfunction); **early prosthesis** easier and preserves length (counsel re infection).
- SCD**: intracavernosal therapy first + concurrent haematology care; do not delay.

Non-ischaemic

- Observation** first (many resolve); persistent → **selective embolisation** (gelfoam/clot - lower ED); surgical ligation rarely.

Stuttering

- Treat episodes as ischaemic; prevent with **daily PDE5i (phosphodiesterase-5 inhibitor)**, pseudoephedrine, self-injection phenylephrine, GnRH (gonadotropin-releasing hormone) agonists/antiandrogens (not if fertility desired); SCD - hydroxyurea.

Diagram to draw: Penile cross-section - distal vs proximal shunts.

II-3b. Renal tubular acidosis

Type: SAQ (10 marks)

Definition

- **RTA (renal tubular acidosis) = hyperchloraemic, normal AG (anion gap) metabolic acidosis** from impaired HCO_3 reabsorption or H^+ excretion, with preserved GFR (glomerular filtration rate).

Types

	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	↓ H^+ secretion, collecting duct	↓ HCO_3 reabsorption, proximal tubule	Aldosterone deficiency/resistance
Urine pH	>5.5 always	<5.5 once HCO_3 below threshold	<5.5
Serum K	Low	Low	High
Urine citrate	Low	Normal/high	Normal
Stones/nephrocalcinosis	Common (calcium phosphate)	Rare	Absent
Causes	Genetic, Sjogren's, amphotericin, MSK (medullary sponge kidney)	Fanconi, myeloma, acetazolamide	Diabetes, obstructive uropathy , spironolactone
Treatment	K citrate 1-2 mEq/kg/d	Alkali 10-15 mEq/kg/d + K	Fludrocortisone , low K diet

- **Type 3:** rare, carbonic anhydrase II deficiency.
- **Incomplete dRTA (distal RTA):** normal HCO_3 but cannot acidify urine – common in recurrent calcium phosphate stone formers.

Diagnosis

- **UAG (urine anion gap) = $(\text{Na} + \text{K}) - \text{Cl}$: negative** → GI (gastrointestinal) loss/proximal; **positive** → distal/type 4.
- **Ammonium chloride test** (0.1 g/kg): pH fails to fall **<5.5** = dRTA.
- **Furosemide-fludrocortisone test** (40 mg + 1 mg): better tolerated.
- FE (fractional excretion) $\text{HCO}_3 > 15\%$ on bicarbonate loading = proximal.

Urological relevance

- dRTA → **recurrent calcium phosphate stones, nephrocalcinosis**, hypocitraturia.
- **MSK** associated with incomplete dRTA.
- **Obstructive uropathy** (posterior urethral valves, prostatic obstruction) → type 4; improves after relief.
- **Topiramate/acetazolamide** → stones.
- **Urinary diversion** → hyperchloraemic acidosis mimicking RTA.
- K citrate preferred over sodium bicarbonate (sodium ↑ Ca excretion).

Diagram to draw: Nephron – proximal tubule (type 2), α -intercalated cell (type 1), aldosterone site (type 4).

II-3c. Millin's prostatectomy

Type: SAQ (10 marks)

Definition

- **Retropubic extravesical simple prostatectomy** through **transverse anterior capsulotomy**, bladder not opened – **Millin, 1945**. Contrast **Freyer's transvesical**.

Indications

- Gland **>80-100 mL** when HoLEP (holmium laser enucleation of prostate) unavailable; usual BPH (benign prostatic hyperplasia) surgery indications.
- **Best for** large lateral lobes, small/no median lobe.
- **Relative contraindications:** large median lobe, diverticulum/large stones (transvesical preferred), small fibrous gland, prior pelvic surgery/radiation, prostate cancer.

Steps

1. Pfannenstiel; extraperitoneal **space of Retzius**.
2. **Haemostatic sutures** across anterior capsule (Santorini plexus).
3. **Transverse capsulotomy** ~1-2 cm distal to bladder neck.
4. **Finger enucleation** in plane between adenoma and capsule.
5. **Divide urethra at apex under vision** (protects sphincter).
6. **Posterior bladder neck wedge excision** (prevents contracture).
7. Sutures at **5 and 7 o'clock**.
8. 20-22 Fr three-way catheter; **watertight capsule closure**; drain.

Millin vs Freyer

	Millin	Freyer
Bladder	Not opened	Opened
Apex	Under vision – better continence	Blind finger fracture
Haemostasis	Fossa visible	Less visible
Stones/diverticulum, median lobe	Difficult	Ideal
Catheter/stay	Shorter	Longer
Specific risk	Santorini bleeding, osteitis pubis	Bladder spasm, suprapubic fistula

Complications

- **Haemorrhage/transfusion**, clot retention, infection; **retrograde ejaculation** (most), bladder neck contracture, stricture, **osteitis pubis**.

Modern alternatives

- **HoLEP** – size-independent, less bleeding; EAU (European Association of Urology) first choice >80 mL.
- **RASP (robotic-assisted simple prostatectomy)** – transvesical or Millin-type; less blood loss, shorter stay; allows diverticulectomy/stone removal.

Diagram to draw: Sagittal section – capsulotomy below bladder neck, enucleation plane.

II-3d. Energy sources in laparoscopic urology

Type: SAQ (10 marks)

Introduction

- Aims: cutting, coagulation, vessel sealing. ~60 °C - coagulation; ~100 °C - vaporization (cutting); >200 °C - carbonization.

Types

Device	Principle	Feature	Limitation
Monopolar	RF (radiofrequency) current through patient to return pad; cut vs coag waveforms	Cheap, versatile	Greatest lateral spread; stray current
Bipolar	Current between jaws	Near nerves/bowel	Slow; no cutting
Advanced bipolar (LigaSure)	Pressure + impedance-controlled energy	Seals up to 7 mm	Cost; spread ~1-3 mm
Ultrasonic (Harmonic)	Blade ~55.5 kHz, <100 °C	Cut + seal up to 5 mm ; no current through patient	Hot blade
Hybrid (Thunderbeat)	Ultrasonic + bipolar	Seals up to 7 mm	Cost
Argon beam	Monopolar via argon, non-contact	Partial nephrectomy bed	Gas embolism
Lasers , hydro-jet	Holmium, thulium; water jet	Endourology; partial nephrectomy	Smoke; not haemostatic

Mechanisms of electrosurgical injury

- Often unrecognised → delayed bowel perforation. 1. **Insulation failure** - current leaks. 2. **Direct coupling** - electrode touches another metal instrument. 3. **Capacitive coupling** - current induced through intact insulation; **avoid hybrid trocars**. 4. **Thermal spread** - ureter, nerves, bowel. 5. **Return electrode burns** - poor pad contact.

Safety measures

- **AEM (active electrode monitoring)** - shuts generator on insulation failure/capacitive coupling.
- **REM (return electrode contact quality monitoring)**.
- Lowest power, brief activation, tip in view; bipolar near critical structures and with pacemakers.

Recent advances

- Smart generators with real-time impedance feedback; robotic vessel sealers; ultrasonic sealing up to 7 mm.

Diagram to draw: Monopolar vs bipolar circuit; capacitive coupling across hybrid trocar.

II-3e. Role of vascular grafts in urology

Type: SAQ (10 marks)

Introduction

- Grafts replace, bypass or extend vessels when primary repair impossible – transplantation, extended oncological resections.

Types

Type	Examples	Pros	Cons
Autologous vein	Saphenous, gonadal	Low infection, small-vessel patency	Limited length
Autologous artery	Internal iliac, inferior epigastric, Carrel aortic patch	Renal artery reconstruction	Limited availability
Synthetic	PTFE (polytetrafluoroethylene), Dacron	Off-the-shelf; IVC (inferior vena cava), aorta	Infection; not in contaminated field
Biological	Cadaveric vessels; bovine pericardium	Infected fields	Availability

Applications

- Renal transplantation** – multiple arteries: bench side-to-side conjoining or end-to-side implantation; short right vein → gonadal/saphenous/IVC extension; Carrel patch; synthetic aorto-iliac bypass for diseased recipient iliacs.
- Renovascular** – stenosis, **renal artery aneurysm**: aortorenal bypass (saphenous, hypogastric, PTFE); ex vivo bench repair.
- RCC (renal cell carcinoma) with IVC thrombus** – cavotomy + primary closure; **patch cavoplasty** if lumen narrowed >50%; **PTFE IVC replacement** if wall invaded and collaterals inadequate; infrarenal IVC often ligated if collaterals.
- Post-chemotherapy RPLND (retroperitoneal lymph node dissection)** – encased aorta → **Dacron/PTFE interposition graft**.
- Dialysis access** – PTFE AV (arteriovenous) graft.
- Vascular injury** – saphenous/PTFE interposition.

Principles

- Autologous** in contaminated/urine-exposed fields and small vessels; synthetic for large high-flow vessels.
- Tension-free, spatulated, intima-to-intima anastomosis; heparin during clamping; Doppler surveillance.
- Omental wrap isolates synthetic graft from urinary tract/bowel.

Complications

- Thrombosis, stenosis, pseudoaneurysm, **graft infection** (often explantation), graft-enteric/urinary fistula.

Recent advances

- Robotic IVC thrombectomy; human acellular vessel for dialysis access/trauma.

Diagram to draw: Bench reconstruction of two renal arteries.

II-3f. Transrectal ultrasonography

Type: SAQ (10 marks)

Introduction

- Probe **6–10 MHz (commonly 7.5)**, **biplanar** (transverse + sagittal); ↑ frequency = ↑ resolution, ↓ penetration.
- Left lateral; DRE (digital rectal examination) first; transverse base→apex, then sagittal.

Zonal anatomy (McNeal)

Zone	Volume	Echo	Cancers
Peripheral	~70%	Isoechoic	~70%
Transition	~5–10%	Hypoechoic	~20–25%
Central	~25%	Slightly hyperechoic	~5–10%

Measurements

- Volume** = $0.52 \times L \times W \times H$ (length × width × height).
- PSAD (PSA (prostate-specific antigen) density) >0.15** suspicious; used after MRI (magnetic resonance imaging) for PI-RADS (Prostate Imaging Reporting and Data System) 3.
- Cancer: **hypoechoic peripheral lesion** (low specificity).

Indications

- Biopsy guidance**; volume for BPH (benign prostatic hyperplasia) surgery planning.
- Ejaculatory duct obstruction**: seminal vesicles >1.5 cm AP (anteroposterior), midline cysts; vesiculography.
- Prostatic abscess drainage; **brachytherapy**, HIFU (high-intensity focused ultrasound), cryotherapy.

TRUS (transrectal ultrasound) biopsy

- 12-core systematic** (base, mid, apex; medial + lateral) ± targeted; **periprostatic nerve block** (lidocaine at base–seminal vesicle junction).
- Complications: haemospermia (commonest), haematuria, rectal bleeding, **sepsis** (FQ (fluoroquinolone)-resistant *E. coli*).
- Prophylaxis: povidone-iodine rectal cleansing; FQ restricted.

Current shift

- EAU (European Association of Urology) 2024/25: transperineal biopsy recommended** – lower infection; prophylaxis may be omitted.
- Pre-biopsy mpMRI (multiparametric MRI) + targeted biopsy**:
 - PROMIS (2017)**: sensitivity for csPCa (clinically significant prostate cancer) 93% vs TRUS biopsy 48%.
 - PRECISION (2018)**: MRI-targeted detected more csPCa (38% vs 26%), less insignificant cancer.

Recent advances

- Micro-ultrasound (29 MHz)** – OPTIMUM RCT (randomised controlled trial): non-inferior to MRI-targeted biopsy.
- Contrast-enhanced ultrasound, shear-wave elastography.

Diagram to draw: Transverse/sagittal zonal views; 12-core template.

Rapid Revision Sheet - Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. Describe medical management of BPH and role of various surgical modalities in its management. (LAQ)

- IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe.
- 5-ARI: ~25% volume ↓, PSA halved, AUR/surgery ↓ ~50%; MTOPS combination ↓ progression 66%.
- AUA sizes: TUIP ≤30; TURP/PVP/PUL/Rezūm/Aquablation 30–80; HoLEP size-independent; simple prostatectomy >80–100 mL.
- IFIS with tamsulosin; bipolar TURP eliminates TUR syndrome; GOLIATH: PVP non-inferior.

II-2. Discuss metabolic evaluation in urinary calculus disease. (LAQ)

- Two 24-h urines on usual diet, ≥4 weeks after event; creatinine for adequacy.
- Targets: volume >2.5 L, Ca <250/300 mg, oxalate <40 mg, citrate >320 mg, uric acid <750/800 mg.
- dRTA: NH₄Cl load fails to lower pH <5.5; cystinuria: cystine <250 mg/L, pH 7.0–7.5, tiopronin.
- Repeat 24-h urine within 6 months; lumasiran for PH1.

II-3a. Management of priapism (SAQ)

- Ischaemic: pO₂ <30, pCO₂ >60, pH <7.25 – emergency.
- Phenylephrine 100–500 µg/mL, 1 mL q3–5 min up to 1 h.
- Distal shunts (Winter, Ebbehøj, T, Al-Ghorab) before proximal (Quackels, Grayhack).
- Prosthesis for >36 h; high-flow → observe/embolise.

II-3b. Renal tubular acidosis (SAQ)

- Type 1: urine pH >5.5, hypokalaemia, hypocitraturia, CaP stones/nephrocalcinosis.
- Type 4: hyperkalaemia, no stones; obstructive uropathy.
- Positive UAG = distal RTA; NH₄Cl 0.1 g/kg – pH fails to fall <5.5.
- Treat dRTA with potassium citrate.

II-3c. Millin's prostatectomy (SAQ)

- Millin 1945: retropubic, transverse anterior capsulotomy, bladder not opened.
- Apical urethral division under vision; 5 and 7 o'clock sutures; bladder neck wedge.
- Freyer (transvesical) preferred for median lobe, stones, diverticulum.
- HoLEP and robotic simple prostatectomy are modern alternatives for >80–100 mL.

II-3d. Energy sources in laparoscopic urology (SAQ)

- LigaSure: vessels up to 7 mm; Harmonic: up to 5 mm at ~55.5 kHz.
- Stray current injury: insulation failure, direct coupling, capacitive coupling.
- Avoid hybrid trocars; use AEM and REM pads.
- Argon beam – risk of gas embolism.

II-3e. Role of vascular grafts in urology (SAQ)

- Autologous (saphenous, hypogastric, gonadal) for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta.
- Transplant: Carrel patch, bench reconstruction, gonadal/IVC extension of short right vein.
- IVC: patch cavoplasty if narrowed; PTFE replacement if wall invaded.
- Post-chemo RPLND may need aortic graft replacement.

II-3f. Transrectal ultrasonography (SAQ)

- Biplanar probe 6–10 MHz; volume = $0.52 \times L \times W \times H$; PSAD >0.15 suspicious.
- 12-core systematic biopsy with periprostatic block.
- EAU: transperineal preferred (lower sepsis); FQ prophylaxis restricted.
- PROMIS and PRECISION established MRI-first pathway.